

0003 CHROME V8 Engine

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JavaScript Engine :-

- JavaScript Code we write can't be understood by computer.
- "JavaScript Engine" is a program that converts JavaScript code that developers write into machine code that allows a computer to perform specific tasks.
 - JavaScript engine executes JavaScript code.
- JavaScript engines are typically developed by web browser vendors.
- V8 - Open-Source JavaScript Engine developed by Google for Chrome.
- SpiderMonkey - The JavaScript Engine powering Mozilla Firefox.
- JavaScriptCore - Open-Source JavaScript engine developed by Apple for Safari
- Chakra - A JavaScript engine for the original Microsoft - Edge [The latest version of Edge uses V8]

Chrome's V8 engine & Node.js

- Chrome's V8 engine by Google sits at the core of Node.js

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V8 is Google's open source high-performance JavaScript and WebAssembly engine, written in C++. It is used in Chrome and in Node.js, among others.

This documentation is aimed at C++ developers who want to use V8 in their applications, as well as anyone interested in V8's design and performance. This document introduces you to V8, while the remaining documentation shows you how to use V8 in your code and describes some of its design details, as well as providing a set of JavaScript benchmarks for measuring V8's performance.

About V8

V8 implements ECMAScript and WebAssembly, and runs on Windows 7 or later, macOS 10.12+, and Linux systems that use x64, IA-32, or ARM processors. Additional systems (IBM i, AIX) and processors (MIPS, ppc64, s390x) are externally maintained, see ports. V8 can run standalone, or can be embedded into any C++ application.

V8 compiles and executes JavaScript source code, handles memory allocation for objects, and garbage collects objects it no longer needs. V8's stop-the-world, generational, accurate garbage collector is one of the keys to V8's performance.

JavaScript is commonly used for client-side scripting in a browser, being used to manipulate Document Object Model (DOM) objects for example. The DOM is not, however, typically provided by the JavaScript engine but instead by a browser. The same is true of V8 — Google Chrome provides the DOM. V8 does however provide all the data types, operators, objects and functions specified in the ECMA standard.

V8 enables any C++ application to expose its own objects and functions to JavaScript code. It's up to you to decide on the objects and functions you would like to expose to JavaScript.

Documentation overview

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GUI and IDE setup

Compiling on Arm64

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Checking out the V8 source code

This document explains how to check out the V8 source code locally. If you just want to browse the source online, use these links:

browse

browse bleeding edge

changes

Using Git

V8's Git repository is located at https://chromium.googlesource.com/v8/v8.git, with an official mirror on GitHub: https://github.com/v8/v8.

Don't just git clone either of these URLs! If you want to build V8 from your checkout, instead follow the instructions below to get everything set up correctly.

Instructions

1. On Linux or macOS, first install Git and then depot_tools.

On Windows, follow the Chromium instructions (for Googlers, for non-Googlers) to install Visual Studio, Debugging tools for Windows, and depot_tools (which on Windows includes Git).

2. Update depot_tools by executing the following into your terminal/shell. On Windows, this has to be done in the Command Prompt (cmd.exe), as opposed to PowerShell or others.

gclient

3. For push access, you need to setup a .netrc file with your Git password:

1. Go to https://chromium.googlesource.com/new-password and log in with your committer account (usually an @chromium.org account). Note: creating a new password doesn't automatically revoke any previously-created passwords. Please make sure you use the same email as the one set for git config user.email.

2. Have a look at the big, grey box containing shell commands. Paste those lines into your shell.

4. Now, get the V8 source code, including all branches and dependencies:

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github.com

LICENSE.stro... Add LICENSE.v8, LICENSE.stro... 13 years ago

LICENSE.v8 Add LICENSE.v8, LICENSE.stro... 13 years ago

LOONG_OW... [LOONG64] Add LoongArch64 ... 2 years ago

MIPS_OWNE... [mips] Add LiuYu to MIPS_OW... 2 years ago

OWNERS Make ycm_extra_conf owned b... 7 months ago

PPC_OWNERS PPC/S390: cleanup owner files 5 months ago

PRESUBMIT.py [presubmit] Make presubmit ch... 2 months ago

README.md [docs] Change links from old wi... 4 years ago

RISCV_OW... Add Lu Yahan to RISC-V owners... 2 years ago

S390_OWNE... PPC/S390: cleanup owner files 5 months ago

WATCHLISTS Add tests directory to watchlist 4 months ago

WORKSPACE [bazel] Remove zlib dependency 10 months ago

codereviews... Make Gerrit the default code re... 6 years ago

README.md

V8 JavaScript Engine

V8 is Google's open source JavaScript engine.

V8 implements ECMAScript as specified in ECMA-262.

V8 is written in C++ and is used in Google Chrome, the open source browser from Google.

V8 can run standalone, or can be embedded into any C++ application.

V8 Project page: https://v8.dev/docs

Getting the Code

Checkout depot tools, and run

Fetch V8

This will checkout V8 into the directory v8 and fetch all of its dependencies. To stay up to date, run

git pull origin

- By embedding V8 into your own C++ application, you can write C++ code that gets executed when a user writes JavaScript code.

- You can add new features to JavaScript itself.

- Since C++ is great for lower level operations like file handling, database connections and new operations, by embedding V8 into your own C++ program, you have the power to add all of that functionality in JavaScript.

Here C++ program we are talking about is Node.js* [Actually Node.js is lot more than just a C++ program - So many additional stuffs are added to C++ program]

Chrome's V8 Engine Summary

- A JavaScript engine is a program that executes JavaScript code.

- In 2008, Google created its own JavaScript engine V8.

- V8 is written in C++ & can be used independently or can be embedded into other C++ programs.

- That allows us to write our own C++ programs which can do everything that V8 can do & more.

- means, our C++ program can run ECMAScript & additional features that you choose to incorporate.

- e.g. Features that are available in C++ but not available with JavaScript.

✓ mjsmp.
v.v. jmp.